

Kunj Rathod

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Education

University of Utah Aug 2023 – Dec 2026
B.S. in Computer Science GPA: 3.7/4.0, Dean's List
Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Research

Circuit-Level Generalization in LLMs via Sparse Feature Circuits Jan 2026 – Present
University of Utah

- Building automated pipelines to extract sparse feature circuits from default tasks and predict generalization to non-default tasks using attribution graphs and Pathways Discovery; analyzing circuit stability across model families.

Spatial Memory for Embodied Agents & Long-Horizon Web Agents Jan 2026 – Present
University of Utah

- Designing retrieval-augmented spatial memory architectures unifying spatial, temporal, episodic, and semantic memory for long-horizon robot navigation and manipulation.
- Evaluating agent benchmarks (OS Marathon, BrowseComp, Mind2Web) across live and offline settings; investigating natural-language-to-simulation environment generation (SceneSmith) for scalable zero-shot policy evaluation.

Multi-Agent Materials Discovery for Extreme Environments Aug 2025 – Feb 2026
STARS Lab, University of Utah (Collaboration: NASA, Microsoft, U.S. DoD)

- Built a multi-agent, graph-augmented extraction pipeline normalizing material-property data from 1,000+ papers into a physics-aware knowledge graph; constraint-based design engine enforcing temperature, creep, and pressure limits for aerospace material selection with benchmarked extraction accuracy.

Industry Experience

Software Engineering Intern — Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Manage Capacities admin portal in Microsoft Fabric, integrating internal telemetry pipelines to give capacity admins an auditable timeline of SKU changes across large enterprise tenants.
- Defining the data contract with UX and backend sub-teams across 3 time zones; building agentic developer workflows on Microsoft's AI stack to accelerate delivery.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction via pipeline optimization and API caching.
- Owned API design, schema, and AWS infrastructure (Lambda, API Gateway, RDS, DynamoDB, S3, Bedrock); shipped 6 features across 4 sprints and implemented distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ legal documents, improving recall 28% and reducing hallucinations 35%; benchmarked 8 LLM families on F1, latency, and cost and built model routing that cut inference spend.

Projects

HirePilot — Autonomous AI Recruiting Pipeline (Podium AI Hackathon Winner) Mar 2026
Python, Claude API, Twilio, Perplexity API, GitHub API

- Built a 9-agent pipeline (Slack-triggered GitHub sourcing, Perplexity-enriched profiles, live adaptive voice interviews via Twilio + Claude, multi-dimensional scoring with ranked shortlist) and shipped it in 24 hours.

BioGraphRAG — Distributed Biomedical Knowledge Graph System May 2024 – Jan 2025
Python, FastAPI, NebulaGraph, LlamaIndex, Docker, AWS

- Engineered a GraphRAG system over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) with a 40% accuracy gain over embedding-only baselines; optimized graph traversal 3x via caching and node pruning (p95 <500ms).

Languages & Technologies

Languages: Python, C++, Java, Swift, TypeScript, SQL

AI/ML: PyTorch, TensorFlow, JAX, sparse feature circuits, attribution graphs, RAG/GraphRAG, multi-agent systems, LangChain, LlamaIndex

Systems & Cloud: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, Bedrock), Azure, Docker, Kubernetes, FastAPI, gRPC, CI/CD, distributed systems